

Software Engineering in Industrial Practice (SEIP)

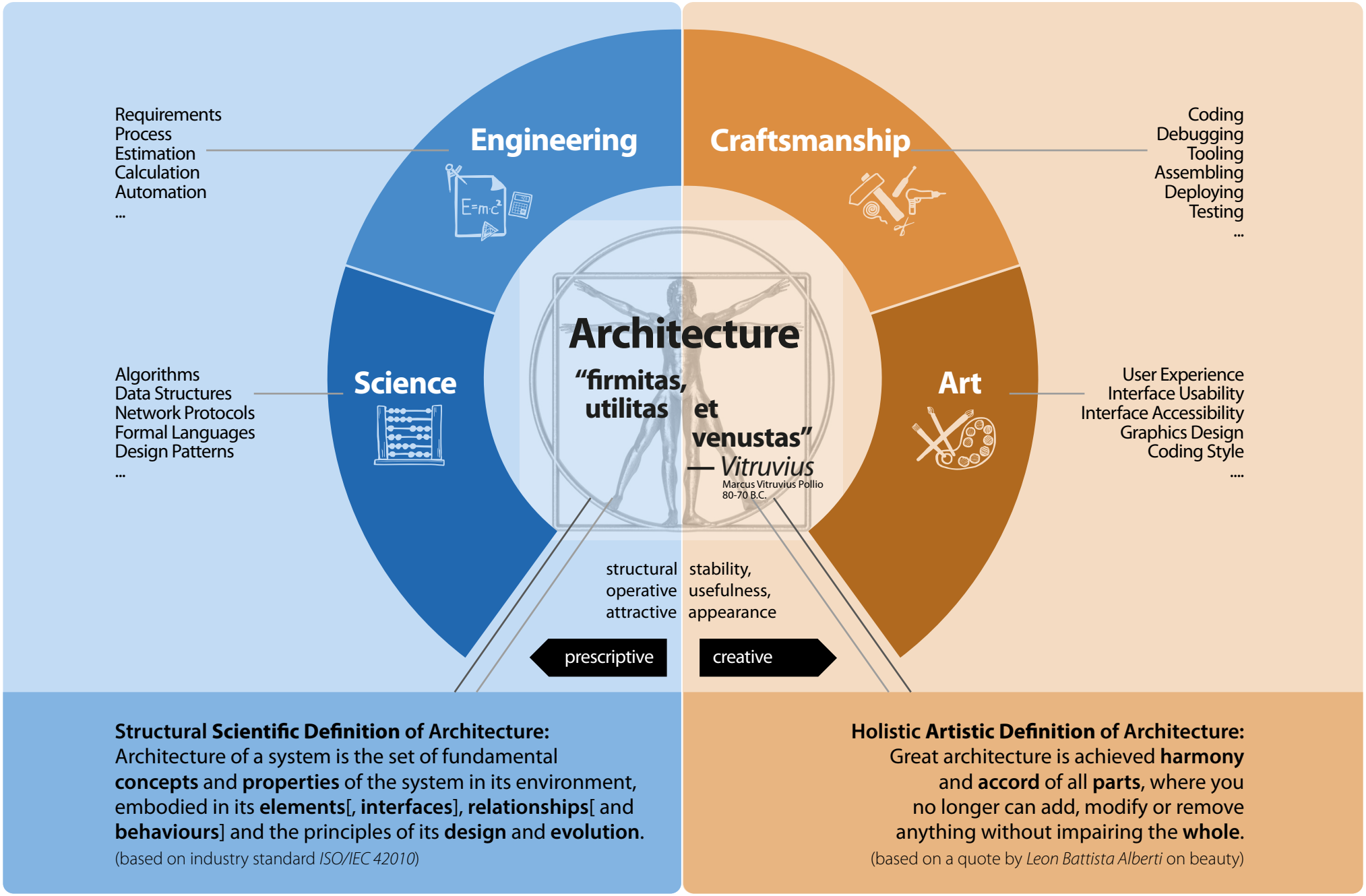
Dr. Ralf S. Engelschall

ARCHITECTURE FUNDAMENTALS

Architecture Stargate

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AF 01.1



Structural Scientific Definition of Architecture:
Architecture of a system is the set of fundamental **concepts** and **properties** of the system in its environment, embodied in its **elements**[, **interfaces**], **relationships**[and **behaviours**] and the principles of its **design** and **evolution**.
(based on industry standard *ISO/IEC 42010*)

Holistic Artistic Definition of Architecture:
Great architecture is achieved **harmony** and **accord** of all **parts**, where you no longer can add, modify or remove anything without impairing the **whole**.
(based on a quote by *Leon Battista Alberti* on beauty)

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Adequacy and Beauty

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AF 01.2

ADEQUACY

1 **Measure**, how **suitable**
2 something,
3 in its environment,
1 fulfills
4 its imposed functional and qualitative requirements (fitness for purpose),
5 from the **individual** and **subjective** perspective of a person.

Beauty does imply **Adequacy**

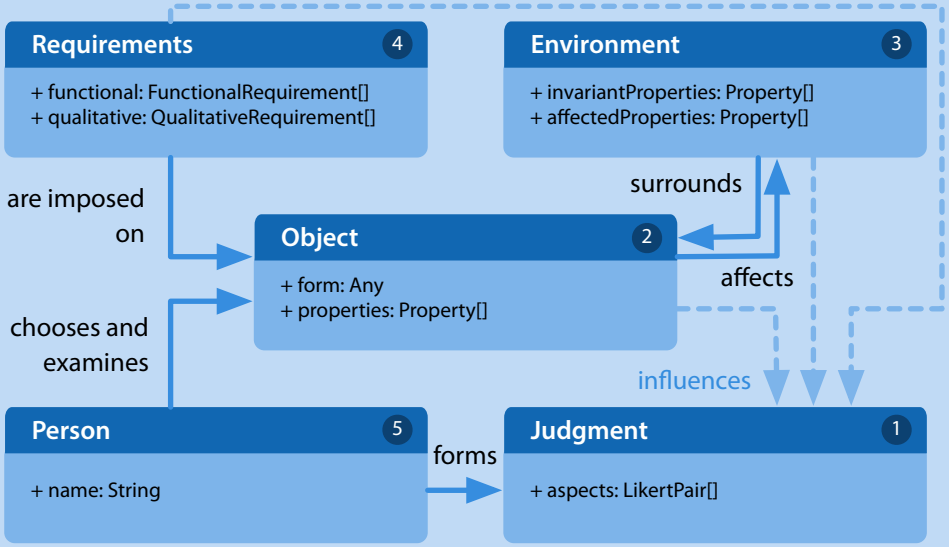
No Beauty does *not* imply No Adequacy

No Adequacy does imply **No Beauty**

Adequacy does *not* imply **Beauty**

BEAUTY

Perception, how **simple, elegant, natural, securing and evident** (SENSE) something is, from the **evolutionary** and **cultural** perspective of a person.



Simple S
free from *unnecessary* complexity, easy *enough* to comprehend.

Elegant E
broadly applicable, allows combinatorial use-cases.

Natural N
consistent, symmetric, free of anomalies.

Securing S
allows *feeling* safe by captured experiences or omnipresence.

Evident E
obvious, clear and traceable representation.

Adequacy is a **relative** and highly **subjective measure** that defies scientific objectification. Nevertheless, it is an essential concept, as the **three influencing aspects** allow persons in practice to at least better **structure** their **perception and judgment** and **measure** the **differences** over time.

Beauty is a **relative** and highly **subjective perception** that defies scientific objectification. Nevertheless, it is an essential concept, as the **SENSE aspects** *seem* to be universal across persons and allow them to **communicate** their **perception and judgment** and **observe** the **differences** over time.

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King Discipline Architecture

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Domain

Many specific Domains

N

Domain
Analytics
Business

domain-specific
usage-oriented
specific

Architecture

One (or few) generic Architectures

1

Architecture
Design
Construction

conceptual
constructional
generic

Technology

Many specific Technologies

M

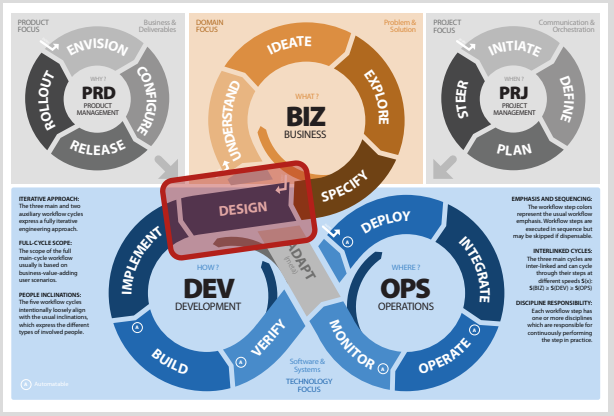
Technology
Development
Realisation

technological
programmatic
specific

Software Engineering Disciplines



Software Engineering Workflow



Architecture Manifesto

Manifesto for IT Architecture

Continuously Raising the Bar

Mission As IT Architects we guide the design, implementation and evolution of IT solutions.

Entitlement We continuously strive to raise the bar of professional IT architecture by practicing it and helping others to learn our craft. We achieve maximum value for our clients through our work.

Values Through this work we have come to value aspects of our craft. While we acknowledge the beneficial values in the items on the right, we appreciate the stronger values in the items on the left even more.

Sustainable Concepts	over	**Latest Technologies**
Pragmatic Making	over	**Theoretical Consideration**
Constructive Craftsmanship	over	**Analytical Engineering**
Accredited Creativity	over	**Achieved Industrialization**
Proactive Improvement	over	**Reactive Correction**
Inherent Quality	over	**Tested Robustness**
Operational Delight	over	**Useful Functionality**



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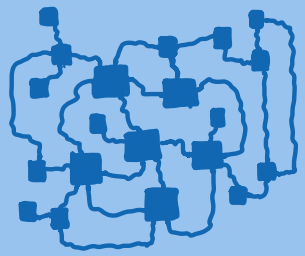
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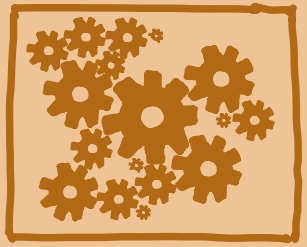
Complex vs. Complicated

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AF 01.5

	complex	complicated
FOCUS	refers to the **extrinsic** and **higher**- or **macro**-level difficulty of a system,	refers to the **intrinsic** and **lower**- or **micro**-level difficulty of a system,
RATIONALE	because the system involves many different and **connected** parts	because the system involves many different and **difficult** aspects
CHALLENGE	which take time to **comprehend** and **master** in **total**,	which take time to **understand** and **learn** in **detail**,
INSIGHT	and which nevertheless are **easy** to explain.	and which usually are **hard** to explain.





NOTICE	NOTICE
Simple (non-complicated) systems can be **complex**.	**Clear** (non-complex) systems can be **complicated**.
RECOGNIZE	**RECOGNIZE**
Architecture primarily has to master the **complex** aspects of a system.	**Development** primarily has to master the **complicated** aspects of a system.

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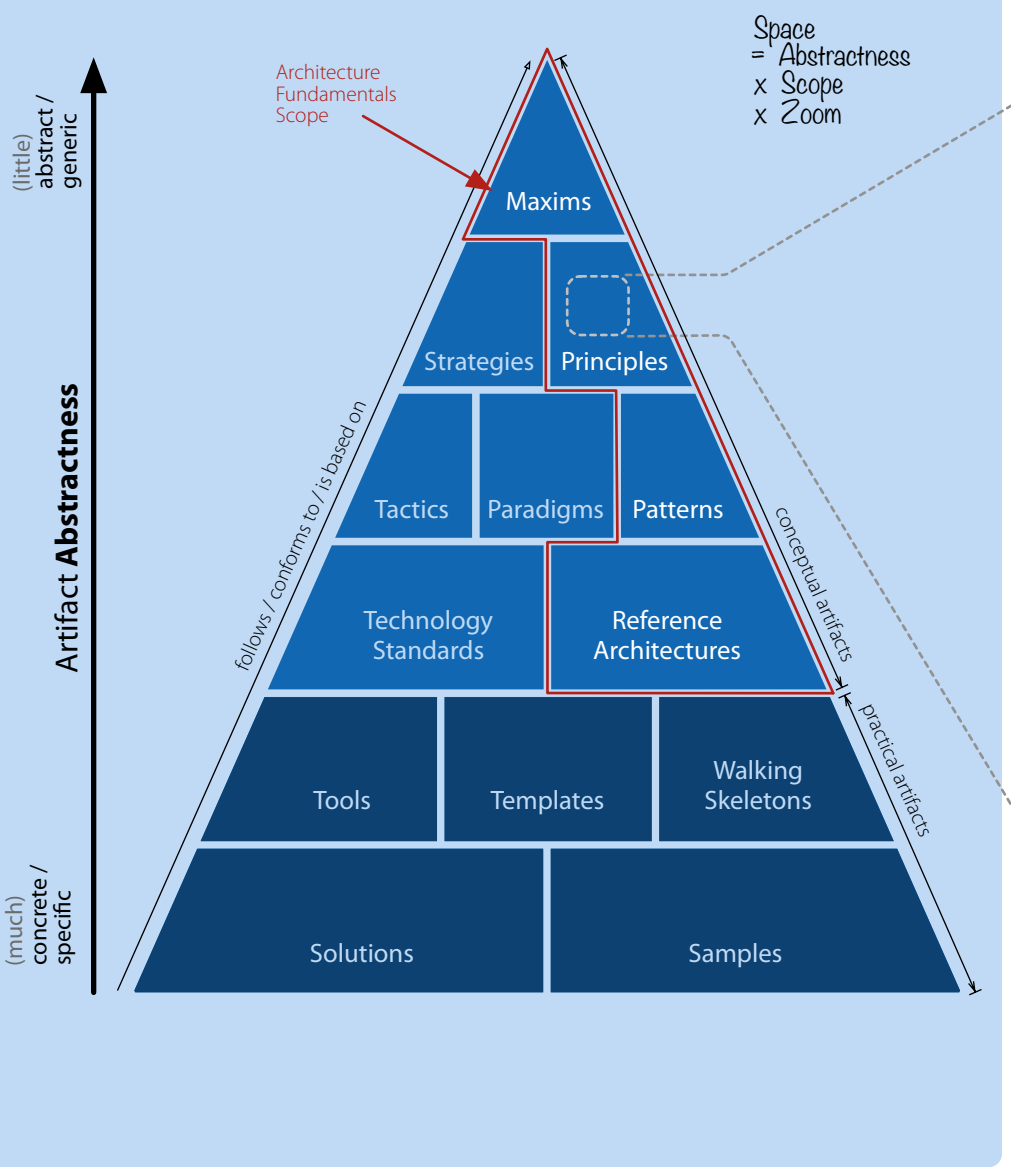
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Architecture Space

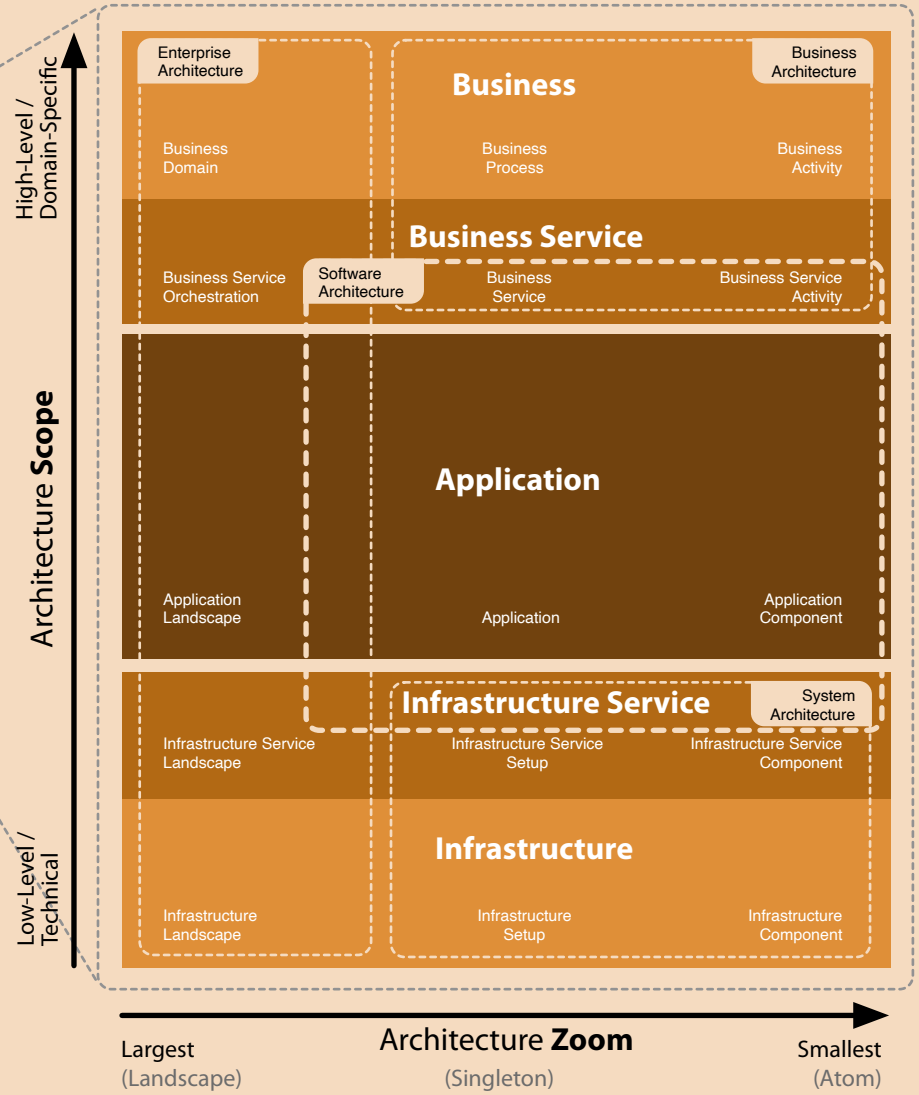
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AF 02.1

Artifact Abstractness



Architecture Scope & Zoom

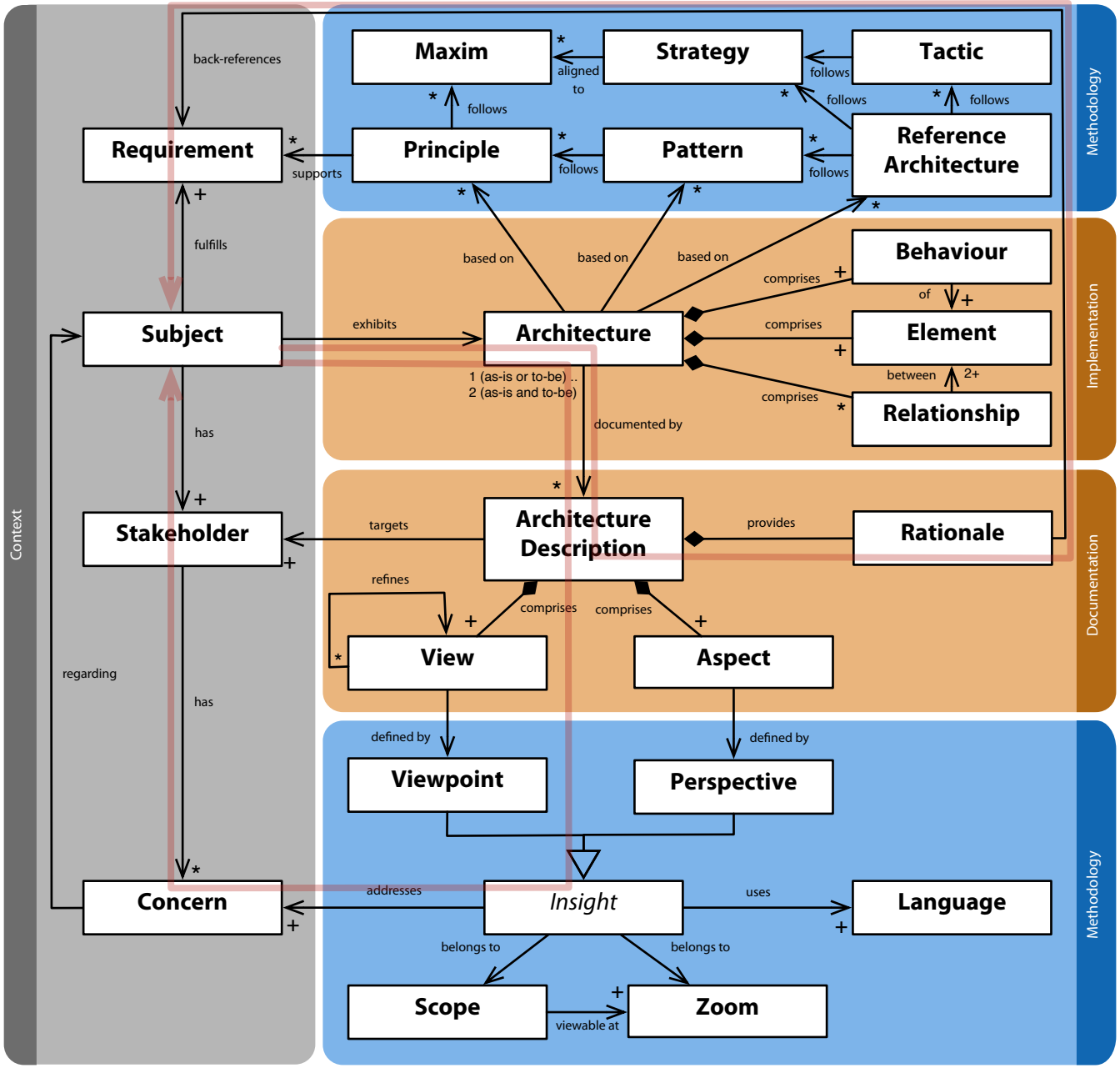


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Architecture Ontology

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Requirement: A functional or non-functional demand or imposed obligation on the **Subject**.

Architecture: Inherent static and dynamic structure of a **Subject** which comprise **Elements**, the visible **Behaviour** of **Elements** and **Relationships** between **Elements**.

Maxim: Fundamental, generally valid set of values and rules to guide the architecture discipline (think: law).

Subject: Any type of business process, business service, software application, infrastructure service or infrastructure setup.

Element: Fundamental part from which a **Subject** can be considered to be constructed.

Strategy: Long-term situation-independent plan of approaches to achieve a particular goal (think: war). Aligned and not in conflict with any **Maxims**.

Stakeholder: Person, group or entity with an interest in or **Concerns** about the **Architecture**.

Behaviour: Run-time characteristic of **Architecture Elements** of the **Subject**.

Tactic: Short-term situation-dependent plan of actions to achieve a particular goal (think: battle). Following and supporting a **Strategy**.

Concern: Requirement, objective, intention, or aspiration a **Stakeholder** has on an **Architecture**.

Relationship: Static or dynamic relationship between **Elements** of an **Architecture** of the **Subject**.

Principle: Fundamental truth, rule, tenet or policy an **Architecture** follows.

Architecture Description: Set of artifacts that document an **Architecture** in a way **Stakeholders** can understand and ensures their **Concerns** are met.

Rationale: Fundamental reasons for a particular chosen **Architecture**, usually strongly based on non-functional **Requirements**.

Pattern: Proven recurring theme, structure, approach or behavior an **Architecture** and its **Elements** can follow.

Viewpoint: Collection of templates and conventions for constructing one type of **View**. It addresses **Concerns** and contains guidelines for constructing a View.

View: Representation of one or more structural aspects of an **Architecture** that illustrates how the **Architecture** addresses one or more **Concerns**.

Reference Architecture: Reusable proven Architecture template, **based** on a set of Patterns and following one or more **Tactics** and **Strategies**.

Perspective: Collection of decisions, guidelines and rules that ensure a **Subject** exhibits a set of non-functional **Requirements**, considered across a number of **Views**.

Aspect: Representation of one or more non-functional aspects of an **Architecture** that illustrate how the **Architecture** addresses one or more **Concerns**.

Scope: Primary area of the **Architecture** space an **Insight** addresses: either Business, Business Service, Application, Infrastructure Service or Infrastructure.

Insight: Superordinate abstract concept to address particular **Concerns** through **Viewpoints** and **Perspectives**.

Language: Formal, semi-formal or even prose language to describe an Insight, so it optimally addresses the **Concerns** of **Stakeholders**.

Zoom: Level of view distance to the **Architecture** an **Insight** has. Also known as the **Insight** detail level, ranging from smallest atoms, over singletons to the largest landscape.

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Requirements Basics

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Requirements Specification

A binding document that specifies the requirements for a solution, by focusing on the WHAT and WHY of the solution — and *not* giving instructions for the HOW.

The documented set of requirements has to be: correct, unambiguous, complete, consistent, ranked, verifiable, modifiable, and traceable.



Requirement Classes

FR Functional (Shall Do)

A condition or capability that a solution must have to provide its service in terms of its behaviour and information. Think: Functionality.



NFR Non-Functional (Shall Be)

A condition, property or quality that a solution must have to satisfy a contract, standard, or other formally imposed obligation. Think: Constraints and "*-ilities".



Requirement Interdependencies

POS Positive (Backing)

One requirement supports the other (e.g. for NFRs: Maintainability and Comprehensibility usually support Adaptability, Portability, Modifiability, etc., and Scalability usually supports Availability, etc.)



NEG Negative (Trade-Off)

One requirement interferes with the other (e.g. for NFRs: Security usually interferes with Efficiency, Usability, Performance, etc., and Orthogonality can interfere with Usability)



Requirement Characteristics

S Specific

The requirement is precise, unambiguous, and clear on what should be done.



M Measurable

The requirement can be verified when it has been achieved by use of a particular test.



A Achievable

The requirement is achievable given existing circumstances and feasible and viable solutions.



R Relevant

The requirement is relevant to the goals of the context.



T Time-Bound

The requirement can be achieved within a reasonable time frame.



Requirement Life-Time

E Enduring

The requirement lasts forever, as it is derived from core activities and organisational structures.



V Volatile

The requirement can be temporary, as it might change over time.



Requirement Expression

[<req-id>] <req-name>:
<subject/actor>
SHALL
<result/action/condition>
BECAUSE
<rationale>



AF 03.1

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Non-Functional Requirements

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AF 03.2

Compliance

CMP Compliance
Ability to meet rules and standards

CRT Certification
Ability to confirm certain characteristics

LCN Licensing
Ability to permit to own and use something

PRC Pricing
Ability to have reasonable price and permit charging for a product

Operation

OPR Operability
Ability to be reasonably operated

SPP Supportability
Ability to be reasonably supported

MNT Maintainability
Ability to cope with changing environments and requirements

TST Testability
Ability to be completely and repeatably tested

TRC Traceability
Ability to track the path something takes

MSR Measurability
Ability to measure characteristics according to defined metrics

Usability

USB Usability
Ability for ease of use, user-friendliness, accessibility, convenience, intuitiveness

CPY Comprehensability
Ability to be easily understood

ACC Accessibility
Ability to be used by people with disabilities.

• • •

Correctness

PRD Predictability
Ability to predict state and behaviour under run-time

FDL Fidelity
Ability to reproduce state and behaviour of the real world

RLV Relevance
Ability to serve as a means to a given purpose

PRN Precision
Ability to be exact and accurate in operation

CRS Correctness
Ability to be algorithmically correct with respect to the specification

PRV Provability
Ability to mathematically prove algorithmical correctness

Protection

SFT Safety
Ability to protect against undeliberate failures, errors and accidents

SEC Security
Ability to protect against deliberate destruction, damage and harm

Availability

AVL Availability
Ability to be operationally available anytime

UBQ Ubiquity
Ability to be operationally present anywhere

RPT Repeatability
Ability to repeat state and behaviour in sequence

RPR Reproducability
Ability to reproduce state and behaviour from scratch

RCV Recoverability
Ability to recover state and behaviour after a disastrous failure

• • •

Quality

RLB Reliability
Ability to perform required functions under stated conditions for a specified time

RSL Resilience
Ability to provide an acceptable level of service in face of faults and challenges

RBS Robustness
Ability to withstand stress, pressure, or changes in procedure or circumstances

STB Stability
Ability to not suffer from internal failures in service

DRB Durability
Ability to keep interfaces and functionality as is for a period of time

INT Integrity
Ability to keep state consistency and avoid data corruption

Performance

PRF Performance
Ability to efficiently perform work, i.e., with a good work to time & resource ratio

SCL Scalability
Ability to scale mostly linearly with changing requirements or conditions

RSP Responsiveness
Ability to respond quickly to external interaction

Structure

SMP Simplicity
Ability to be plain, natural, straight-forward and with no observable complexity

FLX Flexibility
Ability to be easily modifyable in order to respond to altered circumstances

MDL Modularity
Ability to consist of individually comprehensible modules

ORT Orthogonality
Ability to follow great separation of concerns in design

• • •

Execution

EFF Efficiency
Ability to perform work in the most economical way: good input/output ratio

EFC Efficacy
Ability to perform work in order to getting things done and meeting targets

EFV Effectiveness
Ability to perform the "right" work by setting right targets to achieve goals

Interfacing

ITY Interoperability
Ability to correctly operate and exchange information with foreign components

CPT Compatibility
Ability to correctly operate despite expected older or newer interfaces

Evolution

RSB Reusability
Ability to reuse code or data with slight or no modifications

ADP Adaptability
Ability to cope with smaller changes in the run-time environment

PRT Portability
Ability to cope with larger changes in run-time environment

CFG Configurability
Ability to individualize state and behaviour by non-destructive instructions

CST Customizability
Ability to individualize state and behaviour by possibly destructive instructions

EXT Extensibility
Ability to extend state and behaviour in a controlled way

TLR Tailorability
Ability to adjust state and behaviour in a controlled way

MDF Modifiability
Ability to change state and behaviour in an arbitrary way

• • •

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